

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A Debugging & Optimisation Task

Code of Conduct:

I D. Anand Paul (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Anand paul

Questions

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than 1,200 employees use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's Wireless Local Area Network (WLAN). The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use 4G/5G Wireless Wide Area Network (WWAN) services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an Asynchronous Transfer Mode (ATM) network to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size 53-byte cells and establishes Virtual Circuits (VCs) to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak

business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a Multiprotocol Label Switching (MPLS) network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse

terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

5. A leading online education platform conducts live classes and online examinations for more than 15,000 students across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed 10 Gbps connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

1. Problem Identification

The scenario shows two distinct but related networking problems — a WLAN performance/coverage problem inside the building, and a WWAN connectivity/reliability problem for remote/mobile users. Symptoms point to poor RF planning, AP overload, weak roaming support, and outdated wireless security.

2. Root Cause Analysis

Observed Symptom	Likely Root Cause
Slow speed during peak hours	Too many clients associated with a few APs — no load balancing / band steering (2.4GHz vs 5GHz)
Disconnection while moving between floors	Poor roaming support; APs not configured with fast/seamless roaming (no 802.11r/k/v), overlapping cells with mismatched signal strength
Weak signal in conference rooms	Insufficient AP density / poor site survey; RF obstruction from walls, glass, metal fixtures
Video conferencing freezing	No QoS prioritization for real-time traffic; congestion and jitter on shared channels
Uneven AP utilization	Static client association (nearest-AP-only), no controller-based load balancing across APs
WWAN loss in rural/highway areas	Weak cell tower coverage, single-carrier dependency, no failover mechanism
Outdated security protocols	Legacy WEP/WPA still active on some devices instead of WPA2-Enterprise/WPA3

3. Debugging / Diagnostic Steps

- Conduct a wireless site survey (heat-map analysis) floor-by-floor to identify dead zones and overlapping coverage cells.

- Use spectrum analyzers to detect channel interference and overlapping SSIDs on channels 1/6/11 (2.4 GHz band).
- Check AP controller logs for client-count-per-AP to confirm load imbalance.
- Audit authentication logs to list devices still connecting via WEP/WPA instead of WPA2/WPA3.
- Analyze WWAN signal strength (RSSI) logs from field engineer devices along common travel routes.
- Review QoS/DSCP tagging configuration on APs and switches for voice/video traffic.

4. Optimization & Recommended Solutions

Problem Area	Recommended Solution
AP overload / uneven traffic	Deploy a centralized WLAN controller with band-steering and dynamic load balancing across APs
Roaming drops between floors	Enable fast secure roaming (802.11r/k/v) with overlapping cells at -67 dBm design threshold
Weak signal in conference rooms	Add dedicated APs / increase AP density based on re-survey; use directional antennas where needed
Video conferencing quality	Implement QoS with DSCP/WMM prioritization for VoIP and video traffic classes
Channel interference	Reassign non-overlapping channels (1,6,11) and enable auto channel/power optimization (RRM)
WWAN reliability for field staff	Use dual-SIM/multi-carrier routers with automatic failover; deploy SD-WAN for seamless WWAN/WLAN handoff
Outdated security protocols	Migrate all devices to WPA3-Enterprise / 802.1X with RADIUS authentication; disable legacy WEP/WPA

5. Conclusion

The connectivity issues stem from inadequate wireless planning (AP density, roaming, channel allocation) rather than a single point of failure. A combination of a proper RF site survey, controller-based load

balancing, QoS-enabled traffic prioritization, WPA3 migration, and multi-carrier WWAN failover will restore reliable, secure, high-performance connectivity across the building and for remote staff.

SUMMARY — KEY OPTIMIZATION TAKEAWAYS

- Deploy WLAN controller for load balancing + band steering
- Enable fast roaming (802.11r/k/v) and re-survey AP placement
- Apply QoS/DSCP prioritization for VoIP & video traffic
- Upgrade all devices to WPA3-Enterprise / 802.1X
- Use multi-carrier SD-WAN failover for WWAN field connectivity

2.

1. Problem Identification

The core issue is inefficient QoS/traffic-class management in the ATM network. Applications with strict delay/jitter requirements (VoIP, video) are not being served appropriate ATM service categories, causing Virtual Circuit congestion and degraded Cell Loss Ratio (CLR) and Cell Delay Variation (CDV) during peak load.

2. Root Cause Analysis

Observed Symptom	Likely Root Cause
Delayed online transactions	Financial transaction VCs mapped to a lower-priority service category (e.g., UBR) instead of VBR-nrt
Voice jitter / packet loss	VoIP not assigned CBR or rt-VBR category; no strict CDV/CTD bound enforced
Reduced video conferencing quality	Video traffic sharing VCs with bursty best-effort traffic without shaping

Observed Symptom	Likely Root Cause
High switch utilization	No traffic engineering — most traffic routed over few switches/paths (static VC routing)
Congested VCs, others underused	Static VC provisioning not adjusted to actual traffic demand; no PNNI-based dynamic routing
Inefficient bandwidth allocation	Absence of traffic shaping/policing (leaky bucket) at UNI/NNI, allowing bursts beyond negotiated PCR/SCR

3. Debugging / Diagnostic Steps

- Review current ATM service-category mapping per application: CBR, rt-VBR, nrt-VBR, ABR, UBR.
- Check Cell Loss Ratio (CLR) and Cell Delay Variation (CDV) statistics per VC during peak hours.
- Inspect switch buffer occupancy and queue length reports for congested nodes.
- Verify PCR (Peak Cell Rate), SCR (Sustainable Cell Rate) and MBS (Maximum Burst Size) parameters configured per VC against actual traffic profile.
- Confirm whether PNNI routing is enabled for dynamic path selection, or if VCs are statically pinned.

4. Optimization & Recommended Solutions

Problem Area	Recommended Solution
Voice/video traffic quality	Reassign VoIP to CBR and interactive video to rt-VBR with tight CDV bounds
Banking transaction delay	Map transaction traffic to nrt-VBR with guaranteed minimum bandwidth
Best-effort / bulk data	Keep on UBR/ABR so it does not compete with delay-sensitive classes

Problem Area	Recommended Solution
Bursty traffic overloading VCs	Enforce traffic shaping and policing (leaky-bucket/token-bucket) at ingress per PCR/SCR
Congested switches / uneven load	Enable PNNI dynamic routing so VCs are rerouted around congested nodes automatically
Overall capacity shortfall	Increase trunk bandwidth on chronically congested links; add parallel VCs for load sharing

5. Conclusion

Correct ATM service-category assignment combined with enforced traffic shaping/policing and PNNI-based dynamic routing will restore the QoS guarantees ATM is designed to provide, eliminating jitter, packet loss, and transaction delay during peak load.

SUMMARY — KEY OPTIMIZATION TAKEAWAYS

- Map traffic to correct ATM class: CBR (voice), rt-VBR (video), nrt-VBR (transactions), UBR (bulk data)
- Enforce leaky-bucket traffic shaping/policing per PCR/SCR/MBS
- Enable PNNI dynamic routing to balance load across switches
- Add capacity/parallel VCs on chronically congested paths

3.

1. Problem Identification

The MPLS backbone is suffering from poor traffic engineering — traffic is not distributed across available Label Switched Paths (LSPs), some routers have incorrect label bindings causing misrouting/latency, and there is no differentiated treatment for real-time (VoIP/video) versus bulk (ERP/inventory) traffic.

2. Root Cause Analysis

Observed Symptom	Likely Root Cause
Slow cloud application access	LSPs to data centers are congested while alternate paths remain idle — no MPLS-TE load distribution
Delayed order processing	ERP/order traffic sharing best-effort LSP with no bandwidth reservation
Poor voice quality / video interruption	VoIP and video not mapped to low-latency, high-priority LSPs (no DiffServ-aware TE)
Congested vs underutilized LSPs	Static/IGP shortest-path routing used instead of constraint-based routing (CSPF)
Incorrect label mappings	Misconfigured LDP sessions / stale label bindings on some LSRs causing incorrect label swapping
Uneven bandwidth utilization	No RSVP-TE bandwidth reservation or Fast Reroute (FRR) configured for backbone links

3. Debugging / Diagnostic Steps

- Verify LDP neighbor sessions and label bindings on each LSR (show mpls ldp bindings / neighbors).
- Trace label-switched paths end-to-end (traceroute mpls) to confirm consistent label swapping.
- Check per-LSP utilization statistics to identify congested vs idle paths.
- Confirm whether MPLS-TE / RSVP-TE and CSPF are enabled, or if routing is purely IGP shortest-path based.

- Review FEC (Forwarding Equivalence Class) classification to ensure voice/video/data are separated into different traffic classes.

4. Optimization & Recommended Solutions

Problem Area	Recommended Solution
Uneven LSP utilization	Enable MPLS-TE with CSPF (constraint-based routing) to distribute traffic across all available paths
Real-time traffic quality	Implement DiffServ-aware TE: assign VoIP/video to EF/AF classes on low-latency LSPs
Congestion during peak load	Use RSVP-TE bandwidth reservation so critical LSPs are guaranteed minimum bandwidth
Link/node failure resilience	Configure MPLS Fast Reroute (FRR) for sub-50ms failover on backbone links
Incorrect label mappings	Audit and reset LDP sessions on affected LSRs; verify consistent label distribution mode (DoD/DU)
Order processing delay	Prioritize ERP/order-processing FEC with guaranteed bandwidth LSP separate from bulk data transfer

5. Conclusion

Introducing proper MPLS Traffic Engineering (CSPF + RSVP-TE), correcting label-binding errors, and applying DiffServ-based classification will balance load across the backbone, guarantee bandwidth for critical applications, and restore acceptable latency and voice/video quality.

SUMMARY — KEY OPTIMIZATION TAKEAWAYS

- Enable MPLS-TE with CSPF for load-balanced LSP selection
- Reserve bandwidth via RSVP-TE for critical applications
- Classify traffic with DiffServ (EF for voice, AF for video, BE for bulk data)

- Fix LDP label-binding errors on affected LSRs; enable Fast Reroute

4.

1. Problem Identification

The warehouse network is experiencing undetected transmission errors — meaning the current error-detection mechanism is too weak to reliably catch multi-bit / burst errors caused by electrical interference (from motors, robots) and signal attenuation, leading to silently corrupted data reaching the database.

2. Root Cause Analysis

Observed Symptom	Likely Root Cause
Shipment records with wrong data	Weak error-detection code (e.g., simple parity) fails to catch multi-bit burst errors
Barcode scan failures	Signal attenuation / EMI near motors and conveyor systems corrupting scanner-to-server frames
Inventory mismatch with physical stock	Undetected corrupted frames accepted as valid, updating database with wrong values
Some errors retransmitted, some not	Inconsistent error-detection coverage — CRC/checksum not applied uniformly across all links
More errors during peak/congestion	Buffer overflow drops and collisions on shared wireless channels amid heavy RFID/robot traffic

3. Debugging / Diagnostic Steps

- Identify which error-detection method is currently used on scanner/robot links (parity bit, checksum, or CRC).
- Measure Frame Error Rate (FER) and correlate spikes with known interference sources (motor start-up times, peak congestion windows).
- Check whether corrupted frames are silently accepted (no error flag raised) versus correctly flagged for retransmission.

- Inspect physical layer — cable shielding, RFID antenna placement, and wireless channel congestion in the warehouse.
- Review ARQ (retransmission) configuration to see if it is Stop-and-Wait, Go-Back-N, or Selective Repeat.

4. Optimization & Recommended Solutions

Problem Area	Recommended Solution
Weak error detection	Replace simple parity/checksum with CRC (e.g., CRC-32), which reliably detects burst errors common in EMI-heavy environments
Undetected corruption reaching DB	Apply CRC uniformly on every link segment (scanner→AP→server) with mandatory frame rejection on mismatch
Real-time correction need	Introduce Forward Error Correction (Hamming code) on critical short frames so single-bit errors are corrected without retransmission delay
EMI / signal attenuation	Shield cabling, relocate APs/antennas away from motors, use twisted-pair/fiber where feasible
Congestion-related frame loss	Use Selective Repeat ARQ for efficient retransmission and segment the wireless channel to reduce collisions

5. Conclusion

Upgrading from weak parity-based checking to CRC (for detection) combined with Hamming-code FEC (for correction on critical short frames), along with physical-layer interference mitigation, will significantly reduce undetected corrupted records and restore inventory data accuracy.

SUMMARY — KEY OPTIMIZATION TAKEAWAYS

- Replace parity/checksum with CRC-32 for reliable burst-error detection
- Add Hamming-code FEC for real-time single-bit correction on critical frames

- Shield cabling and reposition APs/antennas away from EMI sources
- Use Selective Repeat ARQ to reduce retransmission overhead under congestion

5.

1. Problem Identification

This is a classic sender–receiver speed mismatch (flow control failure), compounded by a congestion problem from 15,000 simultaneous connections. The server transmits faster than many students' devices/links can consume, overflowing receiver buffers and triggering a retransmission storm that worsens overall throughput.

2. Root Cause Analysis

Observed Symptom	Likely Root Cause
Incomplete downloads / delayed loading	Sender transmits at server-side rate without adjusting to each receiver's advertised window (rwnd)
Receiver buffer overflow / app crashes	Fixed / oversized transmission window ignoring low-bandwidth mobile/broadband clients
Duplicate packets, repeated retransmissions	Premature retransmission timeouts triggered by congestion, not actual loss — compounding traffic
Reduced throughput, high latency at peak	No congestion control scaling down send rate as network-wide load increases (all 15,000 users at once)
Single central server bottleneck	No content distribution / caching layer to offload static content (question papers, images, videos)

3. Debugging / Diagnostic Steps

- Check whether the server honours each connection's TCP advertised receive window (rwnd) or uses a fixed send buffer.
- Monitor congestion window (cwnd) growth pattern — confirm if slow-start/congestion-avoidance is active or disabled.

- Correlate buffer-overflow/crash reports with client bandwidth tier (fiber vs Wi-Fi vs mobile).
- Measure retransmission rate and timeout intervals to check for premature retransmissions during transient congestion.
- Review server architecture for presence of caching/CDN layer versus single-origin delivery to all 15,000 students.

4. Optimization & Recommended Solutions

Problem Area	Recommended Solution
Sender/receiver speed mismatch	Enforce proper TCP sliding-window flow control so the server strictly respects each client's advertised rwnd
Buffer overflow on slow clients	Apply adaptive rate control / adaptive bitrate for video and staggered chunked delivery for large files
Retransmission storm under load	Use modern congestion control (e.g., TCP CUBIC/BBR) with proper RTT-based timeout estimation instead of fixed timers
Central server bottleneck	Deploy CDN edge servers / regional caching so students fetch static content (papers, images) closer to their location
15,000 simultaneous connections	Stagger exam start times in batches, or use load balancers with horizontal auto-scaling to distribute connections
Poor bandwidth utilization	Consider QUIC/HTTP-3 for better congestion control and faster recovery over lossy mobile links

5. Conclusion

The platform needs disciplined flow control (respecting each client's rwnd), modern congestion-control algorithms, and horizontal scaling via CDN/load balancers rather than one central server pushing data uniformly to all clients. This will eliminate buffer overflows, cut retransmissions, and ensure smooth exam delivery even under 15,000-student peak load.

SUMMARY — KEY OPTIMIZATION TAKEAWAYS

- Strictly enforce TCP sliding-window flow control (respect client rwnd)
- Adopt modern congestion control (CUBIC/BBR) with adaptive bitrate delivery
- Offload static content via CDN/edge caching to reduce central server load
- Stagger batches / auto-scale servers to handle 15,000 concurrent users